				6.03×10^4 m s^{-1} < v < 6.64×10^4 m s^{-1}	A1
6	**(a)**				
		(i)	$V = 5.0 \times 10^{-3} \times 1200 = 6.0$ V C1 $I = 6.0 \div 4700 = 0.00128$ A ≈ 1.3 mA or 1.28 mA A1		[2]

		(ii)	resistor: (same pd same resistance hence) no change thermistor: (same pd lower resistance hence) increases B1	[1]
	(b)			
		(i)	Minimum $V_{out} = \frac{2.7}{4.5+2.7}(5.0)$ or Maximum $V_{out} = \frac{2.7}{0.7+2.7}(5.0)$ C1 – An attempt (with substitution) to use the Potential Divider Rule or $V = IR$ Minimum V_{out} = 1.9 or 1.88 V A1 Maximum V_{out} = 4.0 or 3.97 V A1 (minus 1 mark if wrong min/max)	[3]
		(ii)	(according to the potential divider rule,) potential difference across the resistor very small fraction of 5.0 V **or** pd across LDR ≈ 5 V B1 Very low range of V_{out} received **or** $V_{out} \approx 0$ at all illumination level or very small difference in V_{out} B1 Therefore very low precision of illumination level calculated or high relative (fractional/percentage) uncertainty of illumination level calculated or processing instrument/circuit not sensitive enough to detect small changes in V_{out} B1 (accept V_{out} too small to be detected but max 2 marks, cannot score for 2nd mark)	[3]

7		